

Charge Fully, Stay Safe: Nested Multi-Timescale Learning for EV Charging Coordination

Abraheem Rashid, Faisal Iradat, Waseem Iqbal, and Yawar Abbas Bangash

Abstract—Uncoordinated electric-vehicle (EV) charging raises the energy cost of distribution feeders and pushes evening voltages below the service-voltage floor. Pricing, dispatch and voltage regulation act on timescales of hours, minutes and seconds, yet most learning-based coordinators use one decision interval. We propose a three-level nested controller. Level 1 (proximal policy optimization) sets an hourly retail price corridor. Level 2, a deep deterministic policy gradient agent shared by the aggregators, dispatches every 15 min as a learned residual on a deadline-aware cheapest-slot plan, and least-laxity-first allocation makes every dispatch feasible for charger and transformer limits by construction. Level 3 is non-parametric: it models drivers' price acceptance and, every 60 s, corrects voltage with charger reactive power bounded by the inverter apparent-power rating, falling back to curtailment only when that headroom is exhausted. We evaluate on the IEEE 33- and 69-bus feeders with AC power flow, real day-ahead prices, household loads and charging sessions, 50 held-out days and 5 training seeds. At 80% EV penetration the controller cuts energy cost by 30.8% relative to uncoordinated charging with no voltage violations (uncoordinated: 3.99% of steps) and is the cheapest learned controller at every penetration level. At nominal load it costs 4.1% more than a price-aware heuristic that executes the same plan per vehicle; when the feeder is stressed beyond its reactive capacity, it curtails 67% less energy and leaves 64% less undelivered. Attached to the baselines, Level 3 removes their violations up to 100% penetration when load forecasts are accurate.

Index Terms—Deep reinforcement learning, distribution networks, electric vehicle charging, multi-timescale control, reactive power, residual policy learning, voltage regulation.

I. INTRODUCTION

GLOBAL electric-car sales exceeded 17 million in 2024 and are expected to pass 20 million in 2025 [1]. Residential charging concentrates in the evening, when drivers return home and household demand is already at its peak. On weak radial feeders this added load depresses end-of-line voltages toward and below the 0.95 p.u. lower limit of the ANSI C84.1 service range [2]–[4], and it buys energy in the most expensive hours of the day-ahead market. Coordinated charging can avoid both, because most vehicles stay plugged in far longer than they need to charge.

Existing coordinators fall into three families. *Optimization-based* schedulers [3], [5] reach good schedules when the model

and the forecasts are accurate, but they need vehicle-level information over the whole horizon and a network model, and they are usually solved on a single decision interval. *Decentralized* schemes such as time-of-use (TOU) tariffs, game-theoretic demand response and priority rules [6]–[8] scale well but cannot see the network. *Deep reinforcement learning* (DRL) [9]–[17] learns from data, but most agents are flat: one policy acts at one rate and voltage limits enter as reward penalties, which give no guarantee.

The underlying problem is a timescale mismatch. Wholesale prices change every hour, aggregators re-dispatch every few minutes, and voltages respond within seconds. *Nested learning* [18] views a learning system as a set of interconnected optimization problems, each updated at its own frequency. We use it as a design principle: one level per timescale, with set points passed down and measurements passed up.

This paper makes four contributions.

- 1) **A three-level nested controller** (Section IV) that matches each level to its timescale: an hourly PPO price-corridor policy, a 15 min DDPG dispatch policy shared by all aggregators, and a non-parametric layer that resolves price acceptance every 15 min and voltage every 60 s.
- 2) **Plan-residual dispatch with feasibility by construction.** Level 2 learns a correction to a deadline-aware cheapest-slot plan (residual policy learning [19], [20]), so training starts from a sound schedule instead of from scratch. Least-laxity-first allocation [8] with a deadline guard maps the aggregate set point to per-vehicle rates that satisfy charger and transformer limits exactly.
- 3) **Inverter-rating-based reactive correction.** Level 3 bounds each charger's reactive power by $Q_i^{\max} = \sqrt{S_i^2 - P_i^2}$ [21], so curtailing active power never removes reactive headroom. A measured-sensitivity Newton step closes the voltage gap, and curtailment is used only when headroom is exhausted. The layer is independent of the policy above it, and we evaluate it on every baseline as well.
- 4) **A reproducible, paired evaluation** (Sections III and V) with AC power flow on two IEEE feeders; real ENTSO-E prices [22], Pecan Street household loads [23] and ACN-Data sessions [24]; 50 held-out days; 5 training seeds; and eight baselines, including a perfect-foresight model-predictive LP-OPF, PPO-Lagrangian, CPO and a goal-conditioned hierarchical agent. One command regenerates every number, table and figure in this paper from the public code.

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II. RELATED WORK

Optimization. Clement-Nyns *et al.* [3] schedule plug-in vehicles with quadratic and dynamic programming to limit losses and voltage deviations, and Ortega-Vazquez [5] adds battery degradation and price uncertainty at household level. Such schedulers need forecasts of arrivals, departures and energy needs; we include a model-predictive LP-OPF with *perfect* foresight as a strong reference.

Decentralized coordination. Mohsenian-Rad *et al.* [6] formulate demand-side management as a game, and Ma *et al.* [7] show that decentralized charging control can fill the overnight valley. Xu *et al.* [8] show that schedules can be improved by giving priority to vehicles with less laxity and longer remaining charging time; our allocation layer follows this principle in its least-laxity-first form.

Learning-based charging. Wan *et al.* [9] learn real-time charging with deep Q-learning and price features; Zhang *et al.* [10] use DDPG; Jin and Xu [11] exploit the structure of optimal policies in a distribution network; da Silva *et al.* [12], Park and Moon [14] and Qian *et al.* [13] study multi-agent and navigation settings; Shibl *et al.* [15] add vehicle-to-grid operation and degradation; Tuchnitz *et al.* [16] control a fleet; and Orfanoudakis *et al.* [17] scale RL with graph neural networks. Reviews cover the wider landscape [25]. EV2Gym [26] and SustainGym [27] provide simulation benchmarks and, for this study, the bundled open data. Saner *et al.* [28] propose a cooperative hierarchical multi-agent system for several stations, and Ye *et al.* [29] use multi-agent DRL for local energy markets. Hierarchical RL in general is surveyed in [30].

Safe RL and voltage control. Constrained policy optimization (CPO) [31] and Lagrangian methods treat safety as an expected-cost constraint; reviews of their use in power systems are [32], [33]. Safety-constrained DRL has been applied to Volt-VAR control [34], [35], active distribution networks [36], real-time voltage control with stability guarantees [37] and multi-agent active voltage control [38]. Multi-agent DRL also coordinates Volt-VAR devices [39] and active and reactive power [40], and physics-informed graph representations make voltage control robust to bad measurements [41], in the spirit of physics-informed learning [42]. EV chargers have supported feeder voltage with vehicle-to-grid active power [43], [44], with reactive power [21], [45] and with coordinated active and reactive control [46]; inverter reactive capability is standardized for distributed energy resources in IEEE Std 1547-2018 [47].

Positioning. Table I summarizes the gap: flat, safe and hierarchical RL learn voltage safety through rewards or constraints, and optimization needs a linearized network model and forecasts of all vehicles. Our controller separates the timescales, keeps feasibility out of the learning problem and corrects voltage with physics on the measured state, using only day-ahead prices and the states of the plugged-in vehicles.

TABLE I
POSITIONING BY METHOD CLASS.

Class	Time-scales	Voltage handling	Dispatch limits	Forecasts needed
Optimization [3], [5]	one	linearized	constraints	all
Rules [6], [8]	one	none	by rule	none
Flat DRL [9], [10]	one	penalty	penalty/clip	none
Safe RL [31], [34]	one	exp. cost	penalty/clip	none
Hier. RL [28], [30]	several	learned	varies	none
Nested (ours)	three	physics	by constr.	prices

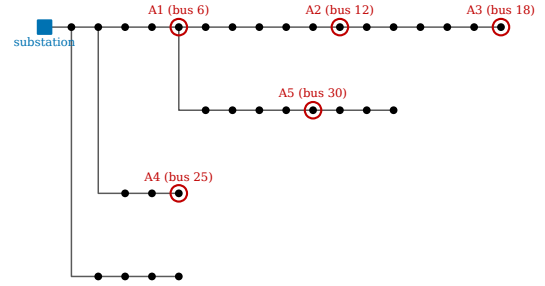


Fig. 1. IEEE 33-bus feeder (MATPOWER data, open ties removed) with the five aggregator connection points.

III. SYSTEM MODEL

A. Feeder and power flow

We use the IEEE 33-bus [48] and 69-bus [49] radial feeders with the MATPOWER data [50] (open tie lines removed). Five aggregators connect at buses 6, 12, 18, 25 and 30 of the 33-bus feeder (9, 21, 35, 49 and 61 on the 69-bus feeder), at electrically strong and weak locations (Fig. 1); vehicles are assigned to them at random in proportions 0.18, 0.18, 0.19, 0.19 and 0.26. Each aggregator serves its vehicles through a 500 kW transformer. The substation voltage is held at 1.03 p.u. throughout. With constant-power loads the radial AC power-flow equations are

$$\mathbf{V} = V_0 \mathbf{1} - \mathbf{Z}^{\text{path}} (\mathbf{S} \oslash \overline{\mathbf{V}}), \quad (1)$$

where $\mathbf{S}$ is the complex load at the non-substation buses, $\oslash$ is element-wise division, the bar denotes complex conjugation, and Z_{nm}^{path} is the impedance of the path shared by buses n and m from the substation. We solve (1) by the backward/forward sweep [51] to 10^{-10} p.u.; our tests match pandapower's Newton-Raphson solution [52] to 10^{-8} p.u., and a power flow is solved every 60 s of simulated time.

B. Data

Prices. Hourly Netherlands day-ahead prices from the ENTSO-E Transparency Platform [22], as distributed with EV2Gym [26], converted to local standard time: 2023 for training, 2024 for testing, and 2019 as an alternative price regime.

Household load. One year of 15 min consumption of 25 Pecan Street homes [23], as distributed with EV2Gym [26], averaged and normalized by its 99.5th percentile. The profile scales the nominal feeder loads, matched by day of year, with

multiplicative noise of standard deviation 0.02 at every power-flow step. The base-load scale is calibrated as the largest value (to 0.01) for which the feeder *without* EVs stays at or above 0.955 p.u. over the whole year (0.78 for the 33-bus and 0.75 for the 69-bus feeder). The voltage problems studied here are therefore caused by EV charging.

Vehicles. The main experiments use a synthetic residential fleet whose parameters are stated assumptions: arrivals $\mathcal{N}(18, 2^2)$ h clipped to 12:00–23:30, dwell $\mathcal{N}(12, 3^2)$ h clipped to 2–16 h, initial and target state of charge $\mathcal{N}(0.35, 0.15^2)$ and $\mathcal{N}(0.85, 0.10^2)$ (clipped to $[0.05, 0.9]$ and to at least 0.05 above the initial value), batteries of 40, 60, 75 or 100 kWh (equally likely), and 11 kW chargers with efficiency 0.95. Episodes run 24 h from 12:00. For generalization we also use real sessions from ACN-Data [24] at Caltech and JPL (bundled with SustainGym [27]): each sampled vehicle keeps a recorded session's arrival time, dwell and delivered energy (2019 for training, 2020–2021 for testing, weekdays, episodes from 06:00). Departures are truncated at the end of the episode and requests capped at what the charger can deliver during the dwell, so full service is attainable except for the wait (under 15 min) for the first dispatch after arrival. EV penetration is the number of vehicles per household, with 300 households behind the five aggregators.

C. Driver response

Each vehicle accepts the execution price p_k of its aggregator with the Fishbein–Ajzen attitude model [53],

$$P_i^{\text{acc}} = \sigma \left(w_i^c \frac{\lambda^{\text{ref}} - p_k}{\lambda^{\text{ref}}} + w_i^n \bar{A} + b_i \right), \quad (2)$$

with logistic σ , reference rate $\lambda^{\text{ref}} = 0.20 \text{ €/kWh}$, last-interval fleet acceptance $\bar{A}$ (floored at 0.3) and heterogeneous weights $w_i^c \sim \mathcal{N}(5, 1)$, $w_i^n \sim \mathcal{N}(2, 0.5^2)$, $b_i \sim \mathcal{N}(0, 0.5^2)$. A driver whose remaining need exceeds 0.7 times the charger rating times the hours to departure accepts regardless of price. Acceptance is drawn once per dispatch interval. The model is stylized; Section VII discusses this limitation.

D. Energy accounting and metrics

A vehicle draws power only while it is plugged in, has accepted the current price and still needs energy, and never more than completes its need within a step. The energy cost is the day-ahead price times the energy actually drawn (after any curtailment). We report the daily cost (€), the cost per delivered kWh, the service quality (SQ, delivered over requested energy), the violation rate (share of 60 s steps with any bus below 0.95 p.u.), the minimum voltage, the curtailed energy, and the mean retail price paid by drivers.

IV. NESTED CONTROLLER

Fig. 2 shows the architecture. Each level is a decision process at its own rate whose environment includes the closed-loop behavior of the faster levels.

A. Level 1: price corridor (PPO, hourly)

The state $s_t^{(1)} \in \mathbb{R}^{11}$ holds the mean bus voltage, the substation active and reactive power, the current day-ahead price, the share of vehicles plugged in, their mean state of charge, the line-loading margin (assumed rating 0.4 kA per line), the substation loading (assumed 8 MVA), the hour (sine and cosine) and the mean price over the next 12 h. The action $a_t^{(1)} \in [0, 1]^2$, drawn from Beta distributions, sets the centre c_t and width w_t of the corridor $[p_t^{\min}, p_t^{\max}] = [c_t - w_t/2, c_t + w_t/2]$ (clipped to $[p^{\text{lo}}, p^{\text{hi}}]$):

$$c_t = \lambda^{\text{ref}} + (2a_{t,1}^{(1)} - 1) \begin{cases} \lambda^{\text{ref}} - p^{\text{lo}}, & a_{t,1}^{(1)} < 0.5, \\ p^{\text{hi}} - \lambda^{\text{ref}}, & \text{otherwise,} \end{cases} \\ w_t = w^{\min} + a_{t,2}^{(1)} (w^{\max} - w^{\min}), \quad (3)$$

with bounds $p^{\text{lo}} = 0.05$, $p^{\text{hi}} = 0.40 \text{ €/kWh}$ and widths $w^{\min} = 0.02$, $w^{\max} = 0.20$. The neutral action $(0.5, 0.5)$ centres the corridor on the reference rate, so an untrained policy, whose mean action is close to it, charges about the flat tariff of the baselines. The reward of hour t ,

$$r_t^{(1)} = -\frac{C_t}{100} - 50(20\Delta V_t)^2 - 10\mathbb{I}[\Delta V_t > 0] \\ - 5\frac{U_t}{100} - 0.5\frac{X_t}{100} + \omega\frac{R_t - \lambda^{\text{ref}}E_t^{\text{del}}}{100}, \quad (4)$$

penalizes energy cost C_t , the voltage deficit $\Delta V_t = \max(0, V_{\min} - V_t^{\min})$, unmet energy U_t of departing vehicles and curtailment X_t ; the last term, with retail revenue R_t and delivered energy E_t^{del} , is the revenue-neutrality term defined below. PPO [54] trains the policy with clipping 0.2, $\gamma = 0.99$ and GAE $\lambda = 0.95$.

B. Level 2: plan-residual dispatch (DDPG, 15 min)

State. The state of aggregator k , $s_{\tau,k}^{(2)} \in \mathbb{R}^{28}$, holds its bus voltage, the hour, the current price, the day-ahead prices of the current and the next 11 hours, the corridor, the number of plugged-in vehicles, their remaining energy need (total, and split into four laxity bins), the power they can absorb, their mean remaining dwell, the last fleet acceptance rate, and the plan set point $u_{\tau,k}^0$ defined below. One actor–critic is shared by the five aggregators, with a one-hot aggregator identifier appended to the state (parameter sharing [55]).

Planning prior. For each plugged-in vehicle i with remaining need e_i , the plan selects the $\lceil e_i/(\eta\bar{c}\Delta) \rceil$ cheapest intervals before departure ($\bar{c} = 11 \text{ kW}$ charger rating, $\eta = 0.95$, $\Delta = 15 \text{ min}$); prices beyond the 12 h look-ahead are valued at the mean known price. The plan set point $u_{\tau,k}^0 \in [0, 1]$ is the power of the vehicles whose current interval is selected, as a fraction of the power the aggregator can deliver now. The plan uses only information the aggregator has: the price look-ahead and the needs and departure times of its plugged-in vehicles.

Action. The actor outputs $a_{\tau,k} \in [0, 1]^2$. The first component corrects the plan,

$$u_{\tau,k} = \text{clip}(u_{\tau,k}^0 + \rho(2a_{\tau,k,1} - 1), 0, 1), \quad (5)$$

so an actor output of $a = 0.5$ reproduces the plan; the residual scale ρ is listed in Table II. The second component sets the

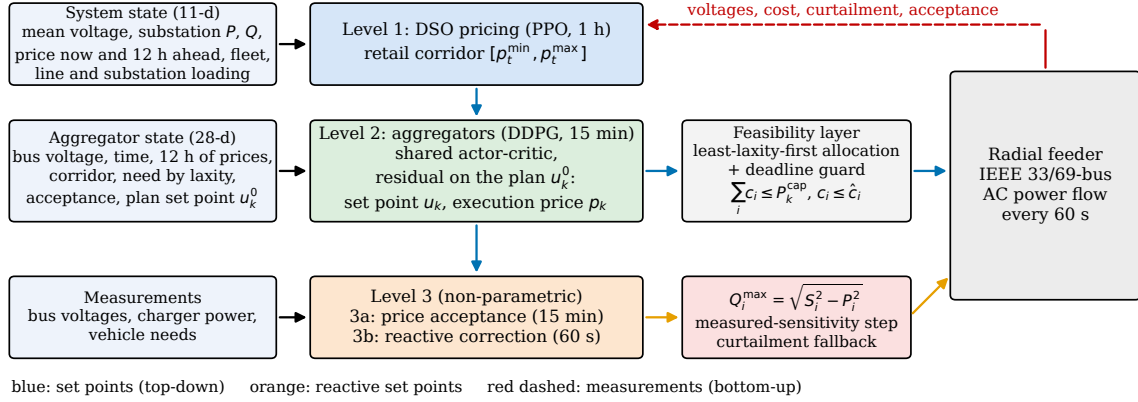


Fig. 2. Nested controller. Set points flow down (blue), Level 3b computes reactive set points from measured voltages (orange), and measurements flow back up (red, dashed).

execution price inside the corridor, $p_{\tau,k} = p_t^{\min} + a_{\tau,k,2}(p_t^{\max} - p_t^{\min})$.

Feasibility by construction. Let $\hat{c}_i = \min\{\bar{c}, e_i/(\eta\Delta)\}$ for the vehicles that accepted the price. The set point dispatches $P_{\tau,k} = u_{\tau,k} \min\{P_k^{\text{cap}}, \sum_i \hat{c}_i\}$, which is filled in order of increasing laxity $\ell_i = h_i - e_i/(\eta\bar{c})$ (h_i : hours to departure) [8]. A deadline guard raises $P_{\tau,k}$ to cover every vehicle with $\ell_i \leq \Delta$, up to the same limit. Every dispatch therefore lies in

$$\mathcal{C}_k = \left\{ c : 0 \leq c_i \leq \hat{c}_i, \sum_i c_i \leq P_k^{\text{cap}} \right\}. \quad (6)$$

Methods that output per-vehicle rates are mapped onto $\mathcal{C}_k$ by exact Euclidean projection (bisection on the dual of the sum constraint).

Reward. The reward of aggregator k in interval τ is

$$r_{\tau,k}^{(2)} = -C_{\tau,k} + \omega(p_{\tau,k} - \lambda^{\text{ref}})E_{\tau,k}^{\text{del}} - 2U_{\tau,k} - 0.5G_{\tau,k} - 0.5X_{\tau,k}, \quad (7)$$

where $C_{\tau,k}$ is the energy cost of the interval (summed over a day it is exactly the daily cost), $U_{\tau,k}$ the unmet energy of departing vehicles, $G_{\tau,k}$ the energy that can no longer be delivered before departure and $X_{\tau,k}$ the curtailed energy (all in kWh, money in €). Because one day spans 96 decisions, we use $\gamma = 0.999$; with $\gamma = 0.99$ a cost incurred 12 h later, e.g. in the cheapest night hours seen from the afternoon, is discounted by almost 40% ($0.99^{48} \approx 0.62$).

Revenue-neutral pricing. A profit-seeking aggregator would raise retail prices, which lowers acceptance and pushes charging into expensive hours. We instead require the mean retail price paid to equal the flat reference rate λ^{ref} charged by the non-pricing baselines. The weight ω in (4)–(7) is the Lagrange multiplier of this constraint. Plain dual ascent oscillated in our experiments, so we use a PI Lagrangian update [56] after every training episode,

$$e = (\lambda^{\text{ref}} - \bar{p})/\lambda^{\text{ref}}, \quad \bar{e} \leftarrow (1 - \alpha)\bar{e} + \alpha e, \\ \omega = \text{clip}\left(K_p \bar{e} + K_i \sum e, \omega^{\min}, \omega^{\max}\right), \quad (8)$$

where $\bar{p}$ is the mean retail price paid in the episode and the integral is clipped against wind-up (gains and bounds in Table II). The multiplier holds the *training-time* average; the final deterministic policy can still sit above or below it. After training we therefore calibrate the tariff level: a constant offset added to every execution price (the sum clipped to $[p^{\text{lo}}, p^{\text{hi}}]$, so it may leave the corridor by up to the offset) is found by bisection such that, on ten randomly drawn 2023 training days, the deterministic policy's mean retail price paid equals λ^{ref} . The same calibration is applied to every learned method, and only training data are used. Prices can thus shift charging in time while drivers pay about the reference rate; the realized retail prices on the test days are reported with the results.

DDPG [57] trains the actor and critic (256–128–64 units) on transitions $(s_\tau, a_\tau, r_\tau, s_{\tau+1})$, where a_τ is the action *actually applied* after the guard, mapped back through (5) and clipped to $[0, 1]$. The actor's output layer is initialized near zero, so training starts from the plan.

C. Level 3: price acceptance and reactive correction

Level 3a is the acceptance model (2). Level 3b runs inside every 60 s step (Fig. 3). After the power flow, if $\min_n V_n < V_{\min} + \delta$ (with $V_{\min} = 0.95$ and $\delta = 0.001$ p.u.), the plugged-in chargers at aggregator bus k offer the headroom

$$\bar{Q}_k = \sum_{i \in k} \sqrt{\max\{S_i^2 - P_i^2, 0\}}, \quad S_i = 12 \text{ kVA}. \quad (9)$$

A charger at $P_i = 0$ thus offers its full rating, and curtailing active power *increases* reactive headroom. The first injection is proportional to the voltage gap (all headroom at a gap of $V_{\min} + \delta - 0.90$ p.u.), weighted toward the most depressed aggregator buses. Each later iteration measures the sensitivity $\hat{\sigma} = \Delta V^{\min}/\Delta Q$ between the last two power-flow solutions and, if both rose, injects

$$\Delta Q = 1.2 (V_{\min} + \delta - V^{\min})/\hat{\sigma}, \quad (10)$$

split across buses by voltage depression and capped by the remaining headroom; otherwise the proportional step is repeated.

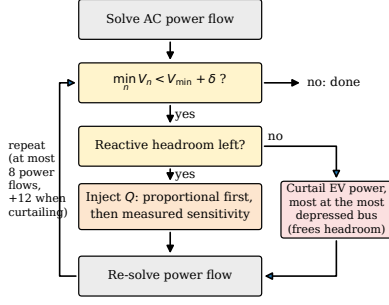


Fig. 3. Level-3b correction inside one 60 s step.

Algorithm 1 One 15 min interval of the nested controller

- 1: **if** hour boundary **then** observe $s^{(1)}$; set corridor by (3)
- 2: **for** each aggregator k **do**
- 3: observe $s_k^{(2)}$ (incl. plan u_k^0); $a_k \leftarrow \mu(s_k^{(2)})$
- 4: set point u_k by (5); execution price p_k
- 5: **end for**
- 6: drivers accept or decline by (2)
- 7: LLF allocation with deadline guard onto $\mathcal{C}_k$ (6)
- 8: **for** each of the fifteen 60 s steps **do**
- 9: solve the AC power flow (1)
- 10: **if** $V^{\min} < V_{\min} + \delta$ **then** Level-3b loop (9)–(10)
- 11: book drawn energy, cost and curtailment
- 12: **end for**
- 13: (training) store $(s, a^{\text{applied}}, r, s')$; update the DDPG agent
- 14: (training, end of day) update the PPO policy and ω by (8)

If no headroom is left, EV power is shed in steps of up to 20% per iteration, most at the most depressed bus, and the flow is re-solved. The loop stops when $V^{\min} \geq V_{\min} + \delta$ or after eight corrective power flows, a limit raised to twenty once curtailment has started. Because the sensitivity is measured on successive AC solutions, the step needs no precomputed network model. Level 3b does not depend on the policy above it; we therefore also evaluate it under every baseline (“+L3”).

D. Training

Episodes are random 2023 days. A curriculum raises the fleet from 60 to 300 vehicles in steps of 60 and advances when the last 20 episodes reach mean SQ ≥ 0.95 with mean curtailment ≤ 20 kWh; the final stage draws the fleet size uniformly from 120–300. Each run has 600 episodes. Algorithm 1 summarizes one dispatch interval, and Table II lists the hyperparameters, read directly from the configuration file.

V. EXPERIMENTAL SETUP**A. Scenarios**

S1, S2, S3 and S5 use 120, 200, 240 and 300 vehicles (40–100% penetration). S4 repeats S3 with a 15% hourly-persistent base-load forecast error (clipped to $\pm 40\%$). Two

TABLE II
HYPERPARAMETERS, READ FROM THE CONFIGURATION FILE OF THE CODE.

Level 1 (PPO)	actor/critic 256–128–64, Beta policy lr 0.0003, γ 0.99, GAE λ 0.95, clip 0.2 10 epochs per update, entropy 0.001
Level 2 (DDPG)	actor/critic 256–128–64, shared, one-hot id lr actor/critic 0.0001/0.001, γ 0.999, τ 0.005 batch 128, buffer 200 000, warm-up 1 000 transitions noise 0.3→0.05 (decay 0.995/episode) residual scale ρ 0.1 PI multiplier: K_p 2.0, K_i 0.05, EMA 0.2, $\omega \in [-2.0, 2.0]$
Training	600 episodes; curriculum window 20
Level 3	δ 0.001 p.u., S_i 12.0 kVA, ≤ 8 (+12 when curtailing) power flows

stress tests probe Level 3: S6 forbids all charging between 19:00 and 20:00 (a failure mode in which reactive capacity tied to charging power would vanish), and S7 has 360 vehicles (120%) with the inverter rating halved to 6 kVA, below the charging power, so a charger drawing more than 6 kW offers no reactive power.

B. Baselines

Rules. Uncoordinated charging (full power on arrival); a TOU timer (off-peak 23:00–07:00, earlier only when laxity is exhausted); and the price-aware heuristic, which charges each vehicle at full power in its planned cheapest intervals, that is, the Level-2 plan executed per vehicle, with the same 12 h price look-ahead. *Model-predictive LP-OPF.* At every interval an LP over the rest of the day (HiGHS [58]) minimizes energy cost plus a 5 €/kWh penalty on unmet energy, subject to per-vehicle energy, charger and transformer limits and to linearized voltage limits ($\geq V_{\min} + \delta$ at every bus) from finite-difference sensitivities of the AC power flow at the forecast base-load (no-EV) operating point of each interval. It has *perfect foresight* of prices and of every vehicle’s arrival, departure and need. It is a strong model-based reference rather than a deployable controller; it is not a strict bound, because its voltage model is linearized and drivers may still decline. *Learned.* Flat DDPG (one agent, all aggregators’ states and actions), PPO-Lagrangian and CPO [31] (expected-cost constraint on violations, limit $d = 0$), and a goal-conditioned hierarchical RL agent (hourly DDPG goals, 15 min DDPG workers). They observe the same states, including the plan set point, use the same allocation layer, reward terms, discount factor and revenue-neutral multiplier, and set execution prices inside the neutral corridor. Because they are trained without Level 3, Flat DDPG and the hierarchical agent add a voltage penalty to the reward (its workers also a goal-tracking term), and PPO-Lagrangian and CPO constrain it as the cost. Each is also evaluated with Level 3 attached. Uncoordinated charging, the price-aware heuristic and the LP-OPF charge the flat reference rate λ^{ref} ; the TOU timer charges its three-tier tariff (0.12, 0.20 and 0.30 €/kWh off-peak, mid-peak and 17:00–21:00).

C. Protocol

All methods are evaluated on the same 50 distinct 2024 days, with identical fleets and noise for each day (paired design). Learned methods are trained with 5 seeds (3 in the ablations and generalization tests); each day's metric is averaged over seeds. We report means with 95% bootstrap confidence intervals over days (10 000 resamples) and compare methods with Wilcoxon signed-rank tests on the paired daily differences, Holm-corrected across the comparison family. On 30 held-out 2023 days and with a criterion fixed in advance (TUNING.md), serving planned slots first lowered the S3 cost by up to 1.3% but left 24–89% (S3) and 39–103% (S7) more energy undelivered, so least-laxity-first allocation was kept.

VI. RESULTS

A. Main comparison

Table III compares all methods at S3. Uncoordinated charging costs 931.4 € per day and violates the voltage floor in 3.99% of the 60s steps. Attaching Level 3 removes every violation of the rule-based methods at S1, S2, S3 and S5 and changes their cost by at most 0.10%, because the correction uses reactive power and curtails only when the headroom is exhausted. The perfect-foresight LP-OPF is the cheapest method (612.0 €). Among the methods that do not see the future, the price-aware heuristic is cheapest (619.8 €). The nested controller costs 645.0 €, which is 30.8% less than uncoordinated charging and 4.1% more than the price-aware heuristic, with no violations, a service quality of 0.997 and a mean retail price of 0.201 €/kWh. It is the cheapest learned controller, 13.0% below the best learned baseline (Hierarchical RL + L3).

B. Penetration and forecast error

Table IV and Fig. 4 extend the comparison to S1–S5. Without Level 3, violations of uncoordinated and TOU charging grow with penetration, up to 6.82% and 7.55%. With accurate load forecasts (S1, S2, S3, S5) the nested controller and every “+L3” method record no violations. S4 is different: with a 15% base-load forecast error, the household load alone breaks the floor on 4 of the 50 days even when no vehicle charges. Level 3 removes most of these violations with the reactive headroom of idle chargers, which offer their full rating at zero charging power; the remainder, at most 0.005% of the steps for the nested controller, occurs only on those days. At every penetration level the nested controller saves 30.6–31.0% relative to uncoordinated charging, costs 3.7–4.3% more than the price-aware heuristic, and stays below every learned baseline by at least 12.6%.

C. Statistical comparison

Table V tests the daily differences at S3. The cost differences to uncoordinated charging and to every learned baseline are significant after Holm correction, and so is the cost premium relative to the price-aware heuristic and the LP-OPF. The “Days” column counts the days on which the nested controller is the cheaper of the pair.

D. Level 3 under stress

Table VI probes Level 3 beyond nominal conditions. In S6 no charging is allowed for an hour in the evening peak; because reactive headroom is bounded by the inverter rating rather than by charging power, every controller with Level 3 keeps the floor. S7 exceeds what halved inverters can correct: even with Level 3 some violations remain, and the methods separate. The nested controller curtails 67% less energy than the price-aware heuristic, leaves 64% less energy undelivered and has 92% fewer violations; its service quality is 0.997 against 0.991. The price of this robustness is energy cost: 982.9 € against 940.4 € per day.

E. What the learned levels add

Table VII separates what each learned level contributes, with Level 3 active throughout. Executed without learning, the plan costs 1.3% more at S3 than the per-vehicle price-aware heuristic, because the aggregate set point is filled least-laxity-first rather than by each vehicle's own cheapest slots. Relative to the plan, learned prices alone raise the S3 cost by 1.0%, learned dispatch alone by 2.8%, and the full controller by 2.7%. The learned dispatch spreads load away from the cheapest slots, which costs energy at nominal load but pays off when the feeder is stressed: in S7 it curtails 38% less energy than the plan and leaves 19% less energy undelivered. Part of the advantage over the price-aware heuristic under stress needs no learning at all: the plan executed least-laxity-first already leaves 65% less energy undelivered than the per-vehicle heuristic in S7, because urgent vehicles are served first when Level 3 curtails.

F. Ablations

Table VIII removes one component at a time (3 training seeds each). The planning prior matters most: learning the dispatch from scratch raises the cost by 7.9% at S3 and by 7.3% at S5. Proportional instead of least-laxity-first allocation raises the S3 cost by 2.3%, a single timescale (Level 1 acting every 15 min) by 0.8%, and removing the deadline guard by 0.9%. Two components do not pay for themselves on cost: with a fixed neutral corridor instead of Level 1 the S3 cost is 0.5% lower, and without the curriculum it is 1.0% lower. We therefore do not attribute a cost benefit to the hourly price level or to the curriculum in this setting. Without the behavioural model every driver accepts, which raises service quality and lowers the cost by 1.0%: price response constrains the controller rather than being exploited by it. Finally, a policy trained without Level 3 violates the floor in 0.23% of the steps, and removing Level 3 from the trained controller at test time gives 0.33%.

G. Generalization

Table IX repeats S3 on the 69-bus feeder, on recorded ACN-Data sessions from two workplace sites, and under the 2019 price regime (the 2019 experiment reuses the policies trained on 2023 prices). In every setting the nested controller

TABLE III

RESULTS AT S3 (240 VEHICLES, 80% PENETRATION; IEEE 33-BUS; 50 HELD-OUT DAYS; LEARNED METHODS AVERAGED OVER 5 TRAINING SEEDS). COST IS THE DAILY ENERGY COST WITH ITS 95% BOOTSTRAP CONFIDENCE INTERVAL; RETAIL IS THE MEAN RETAIL PRICE PAID (€/kWh); SQ IS THE DELIVERED SHARE OF REQUESTED ENERGY. “+L3” ATTACHES THE LEVEL-3 LAYER TO A BASELINE.

Method	Cost (€)	€/kWh	Retail	SQ	Viol. (%)	$V_{\min}$	Curt. (kWh)
Uncoordinated	931.4 [833.8, 1060.7]	0.1146	0.200	1.000	3.99	0.9554	0.0
Uncoordinated + L3	931.4 [835.0, 1062.6]	0.1146	0.200	1.000	0.00	0.9615	5.0
TOU timer	697.7 [639.6, 751.2]	0.0863	0.132	0.995	4.30	0.9372	0.0
TOU timer + L3	697.1 [641.0, 750.9]	0.0862	0.132	0.995	0.00	0.9515	23.7
Price-aware heuristic	619.8 [561.8, 674.2]	0.0766	0.200	0.997	0.73	0.9598	0.0
Price-aware heuristic + L3	619.8 [564.4, 675.0]	0.0766	0.200	0.997	0.00	0.9623	1.7
MPC LP-OPF (perfect foresight)	612.0 [555.7, 667.2]	0.0759	0.200	0.992	0.01	0.9634	0.0
MPC LP-OPF (perfect foresight) + L3	612.0 [555.3, 667.3]	0.0759	0.200	0.992	0.00	0.9635	0.0
Flat DDPG	756.1 [695.5, 828.3]	0.0934	0.203	0.996	0.45	0.9673	0.0
Flat DDPG + L3	756.1 [694.6, 827.0]	0.0934	0.203	0.996	0.00	0.9686	0.1
PPO-Lagrangian	790.2 [725.8, 866.6]	0.0976	0.199	0.996	0.19	0.9758	0.0
PPO-Lagrangian + L3	790.2 [725.2, 865.0]	0.0976	0.199	0.996	0.00	0.9763	0.0
CPO	816.9 [747.5, 900.1]	0.1006	0.200	0.999	0.35	0.9731	0.0
CPO + L3	817.0 [748.1, 903.3]	0.1006	0.200	0.999	0.00	0.9739	0.0
Hierarchical RL	741.3 [685.3, 799.4]	0.0915	0.200	0.997	0.03	0.9741	0.0
Hierarchical RL + L3	741.3 [685.3, 800.0]	0.0915	0.200	0.997	0.00	0.9743	0.0
Nested (proposed)	645.0 [590.0, 700.0]	0.0797	0.201	0.997	0.00	0.9665	0.4

TABLE IV

DAILY ENERGY COST (€) AND VIOLATION RATE (%) ACROSS PENETRATION LEVELS S1–S5 (S4: S3 WITH 15% LOAD-FORECAST ERROR).

Method	S1		S2		S3		S4		S5	
	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.
Uncoordinated	474	0.42	780	2.35	931	3.99	931	4.53	1161	6.82
TOU timer	358	0.06	587	2.39	698	4.30	695	4.31	858	7.55
Uncoordinated + L3	474	0.00	780	0.00	931	0.00	931	0.02	1161	0.00
TOU timer + L3	358	0.00	587	0.00	697	0.00	695	0.00	858	0.00
Price-aware heuristic + L3	317	0.00	520	0.00	620	0.00	618	0.02	773	0.00
MPC LP-OPF (perfect foresight) + L3	313	0.00	513	0.00	612	0.00	610	0.00	765	0.00
Flat DDPG	409	0.09	642	0.30	756	0.45	754	0.84	931	1.11
PPO-Lagrangian	403	0.01	662	0.16	790	0.19	788	0.62	984	0.51
CPO	420	0.03	684	0.27	817	0.35	814	0.93	1016	0.81
Hierarchical RL	380	0.00	622	0.04	741	0.03	739	0.38	922	0.19
Nested (proposed)	329	0.00	540	0.00	645	0.00	642	0.00	806	0.00

TABLE V

PAIRED COMPARISON AT S3: DAILY DIFFERENCES (NESTED MINUS METHOD) WITH 95% BOOTSTRAP INTERVALS AND HOLM-CORRECTED WILCOXON SIGNED-RANK p -VALUES.

Nested vs.	Δ Cost (€)	p_{Holm}	Days	Δ SQ	p_{Holm}
Uncoord.	-286.4 [-379.2, -218.5]	< 0.001	50/50	-0.0028	< 0.001
Uncoord.+L3	-286.4 [-377.8, -219.2]	< 0.001	50/50	-0.0028	< 0.001
TOU	-52.7 [-72.9, -34.2]	< 0.001	41/50	+0.0014	0.059
TOU+L3	-52.0 [-71.9, -34.0]	< 0.001	41/50	+0.0016	0.011
Price-aware	+25.2 [+21.9, +28.7]	< 0.001	0/50	+0.0001	1.000
Price-aware+L3	+25.2 [+21.9, +28.8]	< 0.001	0/50	+0.0001	1.000
LP-OPF*	+33.0 [+28.9, +37.3]	< 0.001	0/50	+0.0046	< 0.001
LP-OPF*+L3	+33.0 [+29.0, +37.3]	< 0.001	0/50	+0.0046	< 0.001
Flat DDPG	-111.1 [-142.0, -89.5]	< 0.001	50/50	+0.0009	< 0.001
Flat DDPG+L3	-111.0 [-141.9, -89.8]	< 0.001	50/50	+0.0009	< 0.001
PPO-Lag.	-145.2 [-179.2, -118.7]	< 0.001	50/50	+0.0011	< 0.001
PPO-Lag.+L3	-145.2 [-179.5, -118.8]	< 0.001	50/50	+0.0011	< 0.001
CPO	-171.9 [-217.5, -137.8]	< 0.001	50/50	-0.0018	< 0.001
CPO+L3	-171.9 [-216.2, -138.7]	< 0.001	50/50	-0.0018	< 0.001
HRL	-96.2 [-108.0, -85.1]	< 0.001	50/50	-0.0001	1.000
HRL+L3	-96.2 [-108.5, -85.5]	< 0.001	50/50	-0.0001	1.000

TABLE VI

LEVEL-3 STRESS TESTS. S6 FORBIDS CHARGING FROM 19:00 TO 20:00; S7 HAS 360 VEHICLES WITH INVERTERS DERATED TO 50%. VIOLATION RATE (%), MINIMUM VOLTAGE (P.U.), CURTAILED AND UNMET ENERGY (kWh PER DAY) AND SERVICE QUALITY.

Method	S6		S7			
	Viol.	$V_{\min}$	Viol.	Curt.	Unmet	SQ
Uncoord.	3.80	0.951	9.00	0	6	1.000
TOU	4.32	0.937	10.62	0	402	0.968
Uncoord.+L3	0.00	0.958	0.35	492	15	0.999
TOU+L3	0.00	0.951	0.17	682	561	0.955
Price-aware+L3	0.00	0.962	0.16	139	106	0.991
LP-OPF*+L3	0.00	0.963	0.00	0	98	0.992
Nested	0.00	0.966	0.01	46	38	0.997

more than the dwell allows at the observed acceptance, so every method delivers a smaller share of the requested energy at Caltech (nested 0.970, heuristic 0.967).

is 14.0–30.8% cheaper than uncoordinated charging and 2.7–6.6% more expensive than the price-aware heuristic, with no violations. Workplace sessions are short and often request

H. Level-3 parameters

Table X varies the correction margin δ and the inverter rating S_i . No setting causes violations or changes the cost

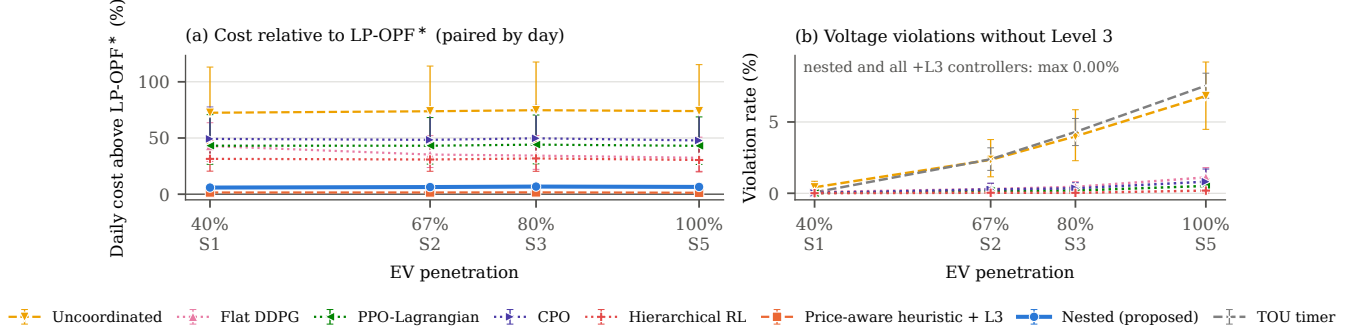


Fig. 4. (a) Daily energy cost relative to the perfect-foresight LP-OPF on the same day, and (b) violation rate of the controllers without Level 3, across S1, S2, S3 and S5 (means with 95% bootstrap confidence intervals over days). The nested controller and every “+L3” controller have no violations at these penetration levels.

TABLE VII

WHAT THE LEARNED LEVELS ADD (LEVEL 3 ACTIVE THROUGHOUT). PLAN: THE LEVEL-2 PLANNING PRIOR EXECUTED WITHOUT LEARNING; PRICES ONLY: LEARNED CORRIDOR AND EXECUTION PRICES WITH PLAN DISPATCH; DISPATCH ONLY: LEARNED RESIDUAL DISPATCH AT THE FLAT TARIFF. DAILY COST (€), RETAIL PRICE (€/kWh), VIOLATIONS (%), CURTAILED AND UNMET ENERGY (kWh PER DAY).

Configuration	S3			S7			
	Cost	Retail	SQ	Cost	Viol.	Curt.	Unmet
Price-aware+L3	619.8	0.200	0.997	940.4	0.16	139	106
Plan+L3	627.9	0.200	0.997	956.7	0.02	114	37
Prices only	633.9	0.201	0.997	966.3	0.03	82	43
Dispatch only	645.3	0.200	0.998	981.4	0.03	71	30
Nested	645.0	0.201	0.997	982.9	0.01	46	38

TABLE VIII

ABLATIONS AT S3 AND S5 (DAILY COST IN €, SQ, VIOLATION RATE IN %).

Configuration	S3			S5		
	Cost	SQ	Viol.	Cost	SQ	Viol.
Complete framework	646	0.997	0.00	807	0.997	0.00
No planning prior	697	0.996	0.00	865	0.996	0.00
No Level 1	643	0.996	0.00	805	0.996	0.00
Single timescale	651	0.996	0.00	813	0.996	0.00
No behaviour model	639	1.000	0.00	800	0.999	0.00
No Level 3 (trained)	648	0.996	0.23	811	0.995	0.71
L3 removed at test	646	0.997	0.33	—	—	—
No curriculum	639	0.996	0.00	797	0.996	0.00
No deadline guard	651	0.996	0.00	815	0.996	0.00
Proportional allocation	661	0.995	0.00	828	0.995	0.00

by more than 0.01%. A larger margin activates the correction more often, and a smaller rating shifts the correction from reactive power to curtailment.

I. Training and computation

Training one nested policy takes about 0.5 h on a single CPU core. The tariff calibration of Section IV moved the learned prices by at most 0.033 €/kWh.

TABLE IX

GENERALIZATION AT 80% PENETRATION: SECOND FEEDER, REAL ACN-DATA SESSIONS AND THE 2019 PRICE REGIME (UNC.: UNCOORDINATED; PA: PRICE-AWARE; LP*: PERFECT-FORESIGHT LP-OPF).

Setting	Daily cost (€)				Viol. (%)		SQ
	Unc.	PA+L3	LP*+L3	Nested	Unc.	Nested	
33-bus, residential	931.4	619.8	612.0	645.0	3.99	0.00	0.997
69-bus, residential	931.4	619.8	611.7	648.0	0.21	0.00	0.997
33-bus, ACN Caltech	146.5	121.3	119.7	126.0	0.00	0.00	0.970
33-bus, ACN JPL	263.8	194.1	197.1	207.0	0.00	0.00	0.994
33-bus, 2019 prices	371.1	282.3	280.2	289.9	4.72	0.00	0.997

TABLE X

SENSITIVITY OF LEVEL 3 AT S3 (NESTED CONTROLLER): CORRECTION MARGIN δ AND INVERTER RATING S_i (DEFAULT $\delta = 0.001$ p.u., $S_i = 12$ kVA).

Setting	Cost (€)	SQ	Viol. (%)	Q act. (%)	Curt. (kWh)
$\delta = 0.0005$	645.0	0.997	0.000	0.30	0.3
Default	645.0	0.997	0.000	0.32	0.4
$\delta = 0.002$	645.0	0.997	0.000	0.37	0.4
$S_i = 10$ kVA	645.0	0.997	0.000	0.32	3.9
$S_i = 14$ kVA	645.0	0.997	0.000	0.32	0.0

VII. DISCUSSION

A. What the learned levels add

At nominal load, learning does not lower the energy bill relative to a well-designed rule: the price-aware heuristic, which executes the Level-2 plan per vehicle, is cheaper than the nested controller, and neither the learned prices nor the learned dispatch reduce cost relative to the plan (Table VII). What learned dispatch provides is a margin: spreading charging away from the very cheapest slots keeps voltages further from the floor and leaves Level 3 more headroom, which matters only on the worst days (S7). It needs the plan as a starting point: learned from scratch, the same dispatch is costlier at every penetration level we tested (Table VIII). We therefore see the nested design not as a way to beat optimization on cost but as a way to keep learned components where they help while planning and physics handle the rest.

B. Scope of the voltage guarantee

Level 3b is corrective control on the simulated quasi-static state, not an a priori guarantee. It restores the floor whenever the reactive headroom of the plugged-in chargers suffices, and otherwise sheds EV power until the floor is met or the fallback has run its course (Section IV). Two consequences follow. First, the inverter apparent-power rating is the design variable that buys voltage margin: Table X reports the effect of S_i , and S7 shows the behaviour when the rating is halved. Second, the correction acts at the 60s resolution of the simulation; switching transients and sub-cycle dynamics are outside the claim, as for every quasi-static study.

C. Assumptions and limitations

Information. The plan, the learned agents and the heuristic know each plugged-in vehicle's departure time and remaining need, as in managed-charging programs where drivers enter them [24]. They do not know future arrivals. Only the LP-OPF reference sees the whole day in advance. *Network.* The feeders are balanced, radial and single-phase equivalent; unbalanced three-phase operation can change the voltage sensitivities on which the correction step relies. *Behaviour.* The acceptance model (2) is stylized. The voltage layer does not depend on it, but the value of price signals does. *Fleets.* The residential fleet is synthetic with stated parameters; the ACN experiments use recorded sessions from workplace sites. *Data.* Household loads (Pecan Street, United States) and day-ahead prices (Netherlands) come from different systems; they provide realistic daily shapes, not a model of one particular utility. Adaptation to structural market change is not tested beyond the 2019 price regime.

VIII. CONCLUSION

We presented a three-level nested controller for EV charging coordination in which each level acts at its own timescale: an hourly price corridor, a 15 min dispatch policy that learns a residual on a deadline-aware cheapest-slot plan, and a non-parametric layer that resolves price acceptance every 15 min and corrects voltage every 60s with inverter reactive power. On two IEEE feeders with AC power flow and real price, load and session data, the controller kept the voltage floor without violations with accurate load forecasts, up to 100% penetration on the 33-bus feeder and at 80% in the other settings, was the cheapest learned controller, and, under a stress test beyond the feeder's reactive capacity, curtailed and left undelivered far less energy than a cost-greedy heuristic. At nominal load that heuristic remains cheaper, and a perfect-foresight LP-OPF cheaper still: the learned levels buy robustness rather than a lower energy bill. Two practical lessons follow. First, a physics-based correction layer bounded by the inverter rating removed the violations of every coordinator we attached it to up to 100% penetration with accurate load forecasts, while changing its energy cost by at most 0.10%, and should be part of every comparison. Second, learned dispatch is worth its complexity mainly when it starts from a sound plan and when grid stress, not price, is the binding concern. Future work includes unbalanced three-phase feeders, uncertainty in departure times, and field validation.

DATA AND CODE AVAILABILITY

The code, configuration, trained policies, evaluation logs and the script that regenerates every table, figure and number of this paper are available at <https://github.com/Abraheem13/nested-ev-grid>. All data are public (Section III); the scripts download the pinned packages that distribute them and verify every file by its SHA-256 checksum.

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